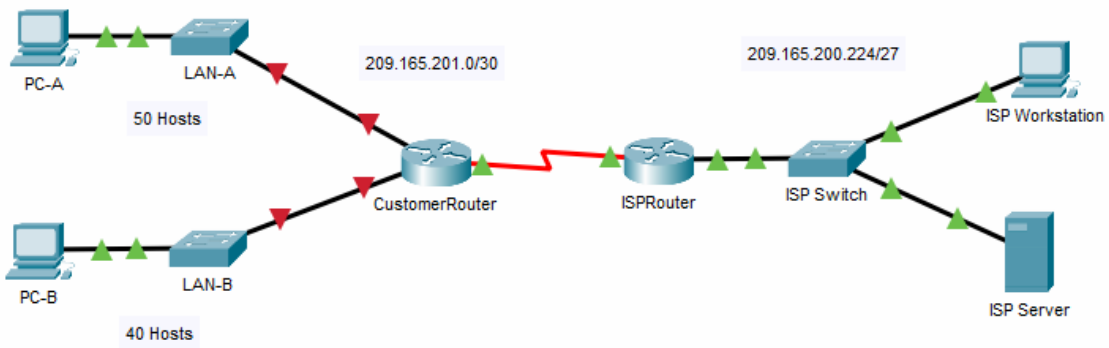
Packet Tracer – Subnet an IPv4 Network

AIM : In this activity, you will subnet the Customer network into multiple subnets. The subnet scheme should be based on the number of host computers required in each subnet, as well as other network considerations, like future network host expansion.

After you have created a subnetting scheme and completed the table by filling in the missing host and interface IP addresses, you will configure the host PCs, switches and router interfaces.

After the network devices and host PCs have been configured, you will use the **ping** command to test for network connectivity



Address table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
CustomerRouter	G0/0			N/A
	G0/1			
	S0/1/0	209.165.201.2	255.255.255.252	
LAN-A Switch	VLAN1			
LAN-B Switch	VLAN1			
PC-A	NIC			
PC-B	NIC			
ISPRouter	G0/0	209.165.200.225	255.255.255.224	N/A
	S0/1/0	209.165.201.1	255.255.255.252	
ISPSwitch	VLAN1	209.165.200.226	255.255.255.224	209.165.200.225
ISP Workstation	NIC	209.165.200.235	255.255.255.224	209.165.200.225
ISP Server	NIC	209.165.200.240	255.255.255.224	209.165.200.225

Part 1: Subnet the Assigned Network

Step 1: Create a subnetting scheme that meets the required number of subnets and required number of host addresses.

In this scenario, you are a network technician assigned to install a new network for a customer. You must create multiple subnets out of the 192.168.0.0/24 network address space to meet the following requirements:

a. The first subnet is the LAN-A network. You need a minimum of 50 host IP addresses.

b. The second subnet is the LAN-B network. You need a minimum of 40 host IP addresses.

c. You also need at least two additional unused subnets for future network expansion.

Note: All of the device subnet masks should be the same length.

d. Answer the following questions to help create a subnetting scheme that meets the stated network requirements:

How many host addresses are needed in the largest required subnet?

What is the minimum number of subnets required?

The network that you are tasked to subnet is 192.168.0.0/24. What is the /24 subnet mask in binary?

e. The subnet mask is made up of two portions, the network portion, and the host portion. This is represented in the binary by the ones and the zeros in the subnet mask.

In the network mask, what do the ones represent?

In the network mask, what do the zeros represent?

f. To subnet a network, bits from the host portion of the original network mask are changed into subnet bits. The number of subnet bits defines the number of subnets.

Given each of the possible subnet masks depicted in the following binary format, how many subnets and how many hosts are created in each example?

2) (/26) 11111111.11111111.11111111.**11**000000

PREFIX	SUBNET MASK	BINARY	NO. OF SUBNETS	NO. OF HOSTS PER SUBNET
/24	255.255.255.0	11111111.11111111.11111111.00000000	2^0=1	2^8-2=>256-2=254
/25	255.255.255.128	11111111.11111111.11111111.10000000	2^1=2	2^7-2=>128-2=126
/26	255.255.255.192	11111111.11111111.11111111.11000000	2^2=4	2^6-2=>64-2=62
/27	255.255.255.224	11111111.11111111.11111111.11100000	2^3=8	2^5-2=>32-2=30
/28	255.255.255.240	11111111.11111111.11111111.11110000	2^4=16	2^4-2=>16-2=14
/29	255.255.255.248	11111111.11111111.11111111.11111000	2^5=32	2^3-2=>8-2=6
/30	255.255.255.252	11111111.11111111.11111111.11111100	2^6=64	2^2-2=>4-2=2

When you have determined which subnet mask meets all of the stated network requirements, derive each of the subnets. List the subnets from first to last in the table. Remember that the first subnet is 192.168.0.0 with the chosen subnet mask.

(/26) 11111111.11111111.11111111.**11**000000 , will provide four subnet . which meets the requiremnet

NETWORK ADDRESS	FIRST USABLE IP ADDRESS	LAST USABLE IP ADDRESS	BROADCAST ADDRESS	NO. OF HOST PER SUBNET	Prefix	Subnet Mask
192.168.0.0	192.168.0.1	192.168.0.62	192.168.0.63	64	/26	255.255.255.192
192.168.0.64	192.168.0.65	192.168.0.126	192.168.0.127	64	/26	255.255.255.192
192.168.0.128	192.168.0.129	192.168.0.190	192.168.0.191	64	/26	255.255.255.192
192.168.0.192	192.168.0.193	192.168.0.254	192.168.0.255	64	/26	255.255.255.192

Device	Interface	IP Address	Subnet Mask	Default Gateway
CustomerRouter	G0/0	*192.168.0.1*	*255.255.255.192*	N/A
	G0/1	*192.168.0.65*	*255.255.255.192*	
	S0/1/0	209.165.201.2	255.255.255.252	
LAN-A Switch	VLAN1	*192.168.0.1*	*255.255.255.192*	*192.168.0.1*
LAN-B Switch	VLAN1	*192.168.0.66*	*255.255.255.192*	*192.168.0.65*
PC-A	NIC	*192.168.0.62*	*255.255.255.192*	*192.168.0.1*
PC-B	NIC	*192.168.0.126*	*255.255.255.192*	*192.168.0.65*
ISPRouter	G0/0	209.165.200.225	255.255.255.224	N/A
	S0/1/0	209.165.201.1	255.255.255.252	
ISPSwitch	VLAN1	209.165.200.226	255.255.255.224	209.165.200.225
ISP Workstation	NIC	209.165.200.235	255.255.255.224	209.165.200.225
ISP Server	NIC	209.165.200.240	255.255.255.224	209.165.200.225